Nagar Yuwak Shikshan Sanstha's

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Aim :

To study and draw UML Use Case diagram for the given case study.

Theory :

What is a Use Case Diagram?

A **Use Case Diagram** is a UML (Unified Modeling Language) diagram that represents the functional requirements of a system.

A UML use case diagram is the primary form of system/software requirements for a new software program under development. Use cases specify the expected behavior (what), and not the exact method of making it happen (how). Use cases once specified can be denoted both textually and visually (i.e. use case diagram).

It does not show the detail of the use cases:

- It only summarizes some of the relationships between use cases, actors, and systems.
- It does not show the order in which steps are performed to achieve the goals of each use case.

Purpose of Use Case Diagram

Use case diagrams are typically developed in the early stage of development and people often apply use case modeling for the following purposes:

- Specify the context of a system
- Capture the requirements of a system
- Validate a systems architecture
- Drive implementation and generate test cases
- Developed by analysts together with domain experts

Notation Description

Actor

- Someone who interacts with the use case (system function).
- Named by a noun.
- An actor plays a role in the business.
- Similar to the concept of a user, but a user can play different roles - e.g. a professor can be an instructor and also a researcher, playing 2 roles with two systems.
- An actor triggers use case(s).
- An actor has a responsibility toward the system (inputs), and expectations from the system (outputs).

Use Case

- Represents a system function (process - automated or manual).
- Named by verb + noun (or noun phrase), e.g. "Do something".
- Each actor must be linked to a use case, while some use cases may not be linked to actors.

Communication Link

- The participation of an actor in a use case is shown by connecting an actor to a use case with a solid link.
- Actors may be connected to use cases by associations, indicating that the actor and the use case communicate with one another using messages.

Boundary of System

- The system boundary is potentially the entire system as defined in the requirements document.
- For large and complex systems, each module may be its own system boundary.
- For example, for an ERP system for an organization, each module such as personnel, payroll, accounting, etc. can form a system boundary for use cases specific to each business function.
- The entire system can span all of these modules, depicting the overall system boundary.

Use Case Relationship

Extends

- Indicates that a specific use case may include (subject to conditions specified in the extension) the behavior specified by a base use case.
- Depicted with a directed arrow having a dotted line. The tip of the arrowhead points to the base use case, and the child (extending) use case is connected at the base of the arrow.
- The stereotype "<<Extend>>" identifies an extend relationship.

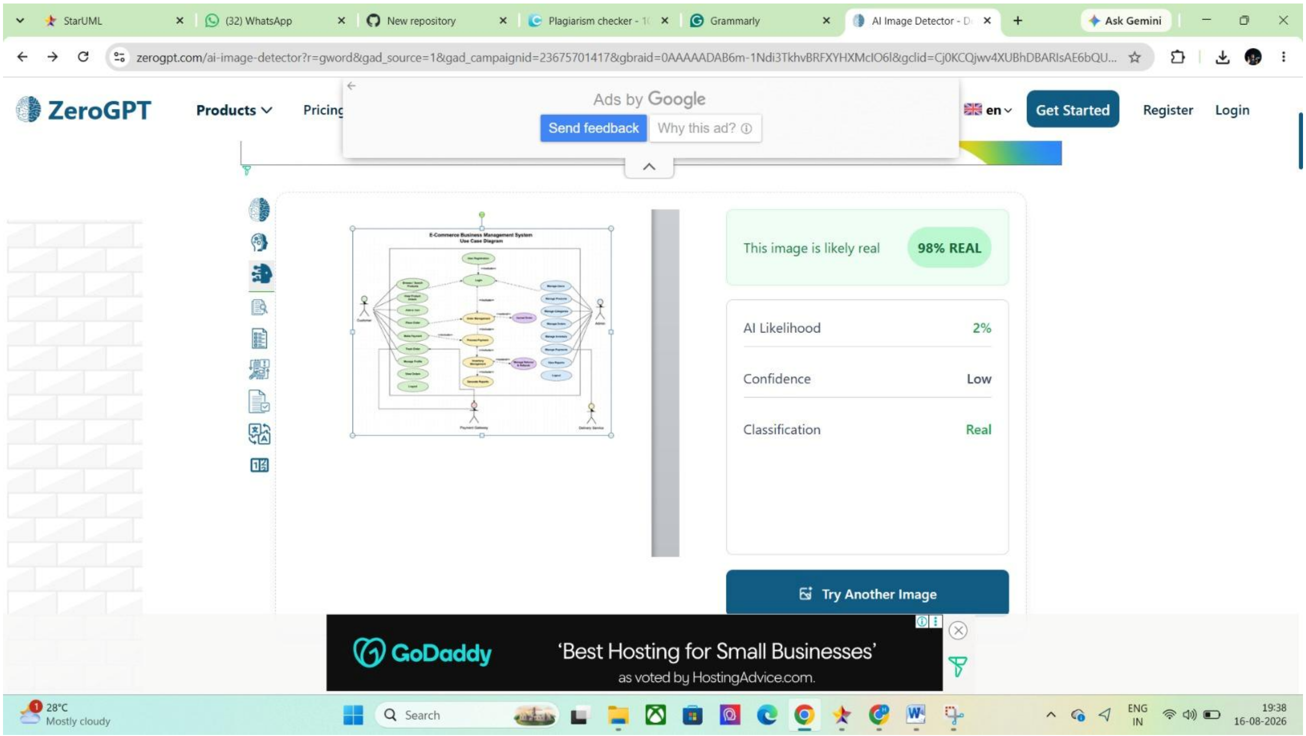
Example (from this case study): The Apply Discount Coupon use case extends the base use case Make Payment - applying a coupon is an optional, conditional step during payment.

Include

- When a use case is depicted as using the functionality of another use case, the relationship between the use cases is called an include (or uses) relationship.
- A use case includes the functionality described in another use case as part of its business process flow.
- An include relationship from a base use case to a child use case indicates that an instance of the base use case will include the behavior specified in the child use case.
- Depicted with a directed arrow having a dotted line. The tip of the arrowhead points to the child use case, and the parent use case is connected at the base of the arrow.

Example (from this case study): The use case Place Order includes the use case Login / Register - a customer must log in before an order can be placed.

- **Plagarism Less than 12 %**



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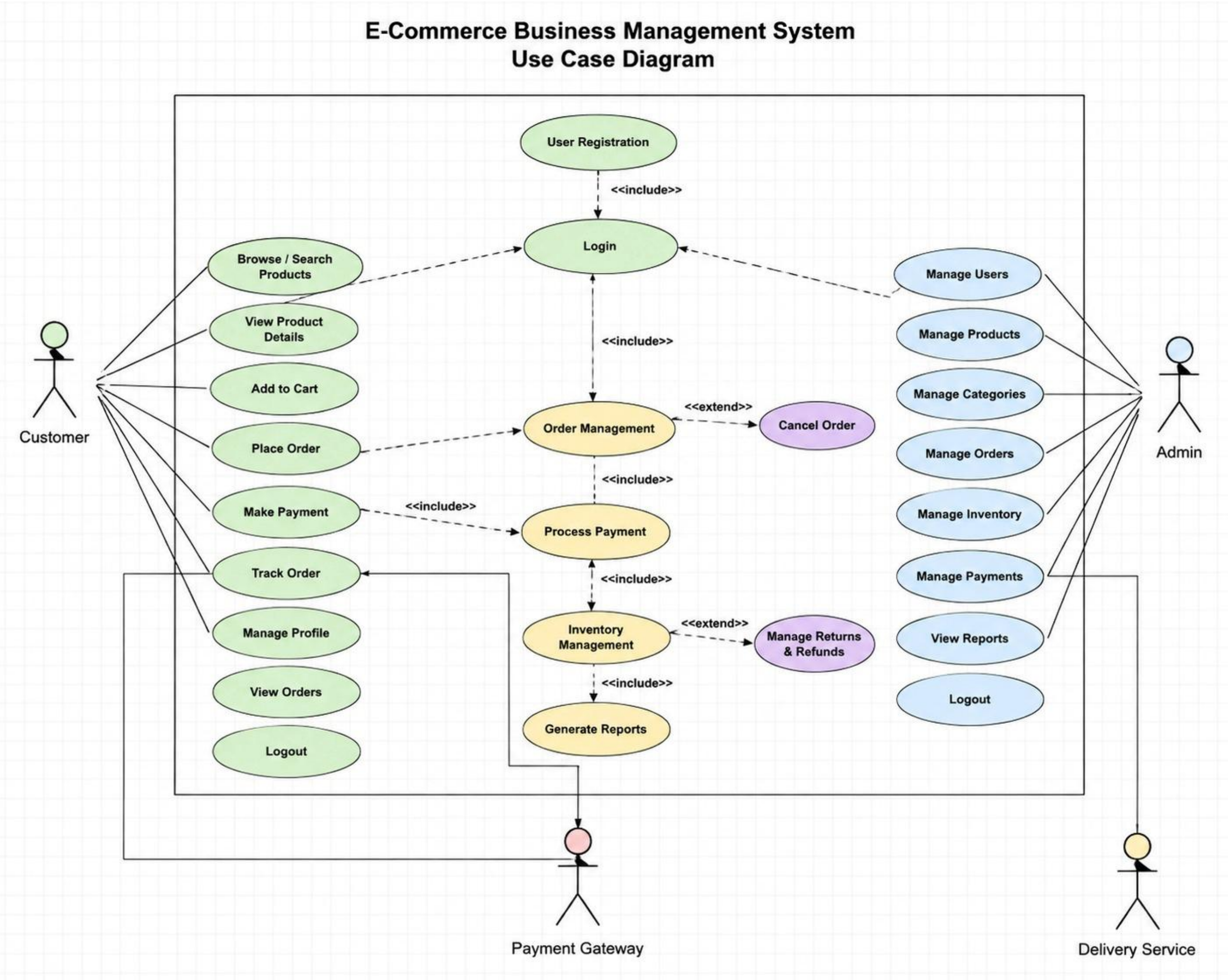
Example : Case Study of E-Commerce Business Management System



Explanation of the Use Case Diagram

System Boundary

The large rectangle labeled "E-Commerce Business Management System" represents the system boundary. All the use cases (functions) provided by the E-Commerce Business Management System are shown inside this boundary.

Actors

There are four actors in the diagram.

1. Customer
2. Admin / Store Manager
3. Delivery Agent
4. Payment Gateway

Use Cases

1. Customer

The Customer (shown on the left) uses the system to browse, shop, and manage their orders.

The customer can:

- Search Products
- View Product Details
- Add to Cart
- Place Order
- Track Order
- View Order History
- Manage Profile
- Make Payment

Placing an order, tracking an order, viewing order history, managing the profile, and making a payment all require the customer to first Log In.

2. Admin / Store Manager

The Admin / Store Manager manages the products, orders, and business reports.

The admin can:

- Manage Products
- Manage Orders
- View Sales Report

These activities also require the admin to Log In.

3. Delivery Agent

The Delivery Agent (shown on the bottom-left) is responsible for order fulfillment.

The delivery agent can:

- Update Delivery Status

This activity also requires the delivery agent to Log In.

4. Payment Gateway

The Payment Gateway is an external system actor responsible for processing transactions.

The payment gateway can:

- Process Payment

The Make Payment use case (triggered by the customer) includes the Process Payment use case, which communicates with the external Payment Gateway.

Main Use Case - Login / Register

The Login / Register use case is placed at the center because every actor must authenticate before accessing the system.

The dashed arrows labeled <<Include>> indicate that each of these use cases includes the Login / Register functionality.

This means:

- Users cannot directly perform any operation.
- They must first log into (or register with) the system.

Included Relationship (<<Include>>)

The <<Include>> relationship means one use case always depends on another.

Examples from the diagram:

- Place Order → <<Include>> → Login / Register
- Track Order → <<Include>> → Login / Register
- View Order History → <<Include>> → Login / Register
- Manage Profile → <<Include>> → Login / Register
- Make Payment → <<Include>> → Login / Register
- Manage Products → <<Include>> → Login / Register
- Manage Orders → <<Include>> → Login / Register
- View Sales Report → <<Include>> → Login / Register
- Update Delivery Status → <<Include>> → Login / Register
- Make Payment → <<Include>> → Process Payment

This shows that authentication is compulsory before using these functions, and that payment processing always relies on the payment gateway's Process Payment use case.

Extended Relationship (<<Extend>>)

The <<Extend>> relationship means a use case optionally extends the behavior of a base use case, under certain conditions.

Example from the diagram:

- Apply Discount Coupon → <<Extend>> → Make Payment

This indicates that applying a discount coupon is an optional step that may extend the Make Payment use case, only when the customer chooses to use a coupon.

Communication Links

The solid lines between actors and use cases are called communication (association) links. They indicate which actor can perform which operation.

For example:

- Customer ─── Search Products
- Customer ─── Place Order
- Customer ─── Make Payment
- Admin / Store Manager ─── Manage Products
- Admin / Store Manager ─── View Sales Report
- Delivery Agent ─── Update Delivery Status
- Payment Gateway ─── Process Payment

Conclusion

The UML Use Case Diagram for the E-Commerce Business Management System was successfully studied and designed. The diagram identifies four actors - Customer, Admin / Store Manager, Delivery Agent, and Payment Gateway - along with their respective use cases. The central Login / Register use case, linked through <<Include>> relationships, ensures that authentication is enforced across the system, while the <<Extend>> relationship for Apply Discount Coupon demonstrates an optional extension to the core Make Payment functionality. This diagram provides a clear, high-level view of the system's functional requirements and the interactions between its actors.